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Experimental and theoretical study of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ ($X = V$ or Cr) refractory high-entropy alloys

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ABSTRACT

We investigated the microstructure and mechanical properties of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ ($X = V$ or Cr) high-entropy alloys (HEA), produced by induction melting and casting in inert atmosphere. The structures of these alloys were studied via X-ray diffractometry and scanning electron microscopy. Results show that $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ has mainly the body centered cubic (BCC) structure, whereas the BCC matrix of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ contains small amount of Cr_2Nb and Cr_2Hf intermetallic compounds. $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy shows the high strength and the homogeneous deformation under compression at room temperature. The strength and hardness of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy are further enhanced by the Cr-containing Laves phases segregated during casting. The structural and mechanical properties remained almost unchanged after a short time (10 min) heat treatment at 573, 773, 973 and 1173 K indicating the resistance to working temperature peaks for these two alloys. Ab initio calculations predict ductile behavior for these and similar refractory HEAs. The theoretically calculated Young's modulus E is in good agreement with the experimental ones.

Introduction

In the high-temperature applications, metallic alloys need to satisfy the superior mechanical and functional properties. A recently proposed concept of high-entropy alloys (HEA) [1], especially consisting of refractory elements [2–5], opens up the possibilities in the alloy developments satisfying the above demands. HEAs are composed of equiatomic or nearly equiatomic metallic elements, wherein the mixing entropy is maximized, leading to the extended solid solubility and formation of a single or a double-phase structure [1,6,7]. Such solid solutions with multi-principal elements have been found to be kinetically stabilized due to the sluggish diffusion of atoms [7–9], preventing intermetallic compounds from precipitating. It is widely accepted that the formation of refractory-HEAs by casting is facilitated by small atomic size difference as well as small and negative enthalpy of mixing between the constituent elements (Hume-Rothery rule [10]), which is opposite to bulk metallic glasses which have relatively large atomic size mismatch and negative mixing enthalpy [11].

Using vacuum arc melting technique, Senkov et al. [2] synthesized near equiatomic-concentration WTaMoNb and WTaMoNbV alloys with single-phase BCC lattice. Since then, many other HEAs based on the refractory elements (Ti, Zr, Hf, V, Nb, Ta, Cr, Mo and W, namely the 4th, 5th, and 6th element groups in the periodic table) have been in the focus of experimental studies [3–5,12–16], for instance TiZrHfNbTa [4,5], $NbZrTiCrMo_{0.5}Ta_{0.5}$ [12], CrNbTiVZr [13,14] and $TiZrNbMoV_x$ ($x = 0$–1.5) [15] as well as TiZrHf [16]. The microstructure and mechanical properties of the above HEAs were investigated at different temperatures. del Grosso et al. [17] employed the Bozzolo–Ferrante–Smith (BFS) method [18,19] to study phase stability of refractory elements.

In order to further enrich and understand the refractory HEAs, we prepare two new equiatomic multi-component HEAs consisting of the 4th and 5th as well as 4th and 6th element groups, having compositions of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$, respectively. We investigate the effects of V and Cr alloying elements on microstructures and mechanical properties of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloys, respectively. As the thermal stability is an important consideration for industrial application of HEAs it is also worthwhile to investigate the annealing effects on the structure and property evolution. In this paper the effects of annealing on the structure and mechanical properties are also investigated.

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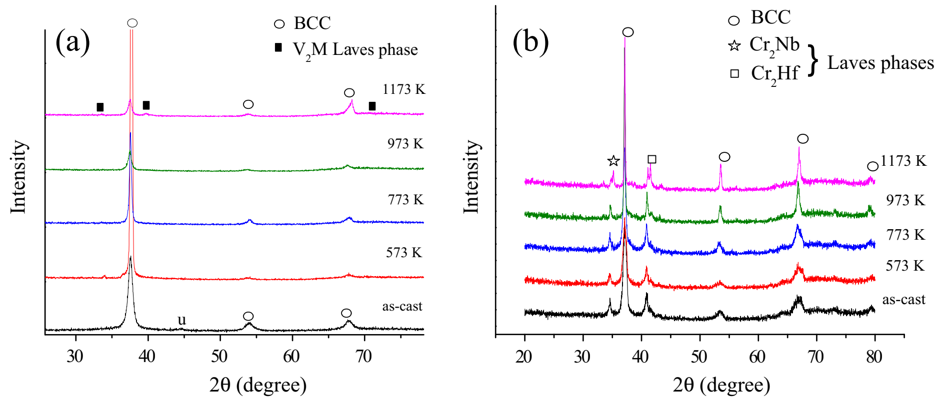

Fig. 1. X-ray diffraction pattern of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ (a) and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ (b) alloys in the as-cast and annealed states, for the V_2M Laves phase, M represents Ti, Zr, Hf, and Nb.

Using the ab initio alloy theory [20], we calculate and compare the structural and elastic properties of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Ta, Cr), $W_{25}Nb_{25}Mo_{25}Ta_{25}$ as well as $W_{20}Nb_{20}Mo_{20}Ta_{20}V_{20}$ and $Zr_{33.3}Hf_{33.3}Nb_{33.3}$ refractory HEAs. It has been proved that ab initio alloy theory provides a powerful tool to understand and predict the behavior of materials. For the present refractory HEAs, the chemical compositions are particularly complicated. The common supercell approach to model these systems becomes very cumbersome due to the larger number of components and chemical disorder. On the other hand, the coherent potential approximation (CPA) [21,22] is considered to be an efficient method for studying the substitutional solid solutions. The CPA in combination with the exact muffin-tin orbitals (EMTO) has been assessed in numerous successful applications [20] and most recently it was used to investigate the phase transitions and the elastic properties of HEAs [23–25].

Methods

Experimental procedure

The ingots with nominal compositions of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ were prepared by induction melting using mixtures of the high-purity elements (>99.9 at.% purity) in an argon atmosphere purified by Ti-getter. Cylindrical rod samples with a diameter of 3 mm were obtained by re-melting the master alloy of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ in a quartz tube and subsequent injection into a copper mold in an inert gas atmosphere.

In the case of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy rectangular-shaped castings were produced. In order to examine the thermal stability of the samples, the isochronal annealing was performed at 573, 773, 973 and 1173 K for 600 s. A high vacuum of 2×10^{-2} Pa was reached and then argon gas was introduced before the annealing.

Table 1
The lattice parameter *a* (pm) of the pure metals and of the BCC phases were presented in the studied $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr) alloys.

Metal	*a*	Metal	*a*
Ti	327.6	V	330.1
Zr	358.2	Nb	302.0
Hf	355.9	Cr	288.4
HEA	*a*(Vegard)	*a*(Exp.)	*a*(DFT)
X = V	334.76	338.5	340.07
X = Cr	332.04	342.9	337.12

(Vegard) represents the values of the BCC phases via Vegard's law, (Exp.) stands for the experimental values from the XRD, (DFT) is the calculated results from ab initio.

The structure of the alloys at cross-sections of the as-cast samples was examined by X-ray diffraction (XRD) with Brucker D8 diffractometer with Cu Kα radiation. The alloy specimens were polished and etched with the mixture of nitric acid (1.40) 30 ml, hydrofluoric acid (40%) 10 ml and water 60 ml. The initial microstructure and the morphologies of the surface of these specimens subjected to mechanical tests were observed using a Hitachi S-4800 Scanning Electron Microscope (SEM). We recorded quantitative chemical analysis using an EDS detector attached to the field emission gun SEM.

In order to evaluate the mechanical properties the cylindrical and rectangular specimens with an aspect length to width ratio of 2:1 were uniaxially compressed at room temperature until fracture (using a Shimadzu testing machine with 50 kN maximum load). The specimens for compression testing were cut from the 3 mm rods and carefully polished to have a final length of 6 mm, while ensuring that their end planes were parallel to each other. The compression test was performed according to the American Society for Testing and Materials (ASTM) standard (E9-09), at an initial loading rate of $5 \times 10^{-4}\ s^{-1}$. The Vickers microhardness was measured on polished cross-section surfaces using a HM211 Mitutoya device. An average of ten measurements has been taken on each sample, on randomly selected places.

Computational methodology

In the present EMTO–CPA calculations, the one-electron equations were treated within the scalar relativistic and frozen core approximations. We employed the local density approximations [26] as the exchange-correlation density functional. The Green's function was calculated for 16 complex energy points around the valence states. The basis set included *s*, *p*, *d*, and *f* states. The irreducible wedge of the BCC Brillouin zone was sampled by 285 inequivalent *k* points. All calculations were performed for static lattice. The singe-crystal elastic constants were determined according to the standard methodology [20] and the polycrystalline elastic moduli were obtained via the Voigt–Reuss–Hill averaging method [27]. We should mention that the present theoretical results correspond to solid solutions and to static conditions, and therefore they are strictly valid at low temperatures and for completely random systems (without significant short range order effects).

Results and discussions

Crystal structure by XRD

Fig. 1(a, b) shows the X-ray diffraction patterns of the as-cast and heat-treated $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloys, respectively. After annealing at 573, 773 and 973 K the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy

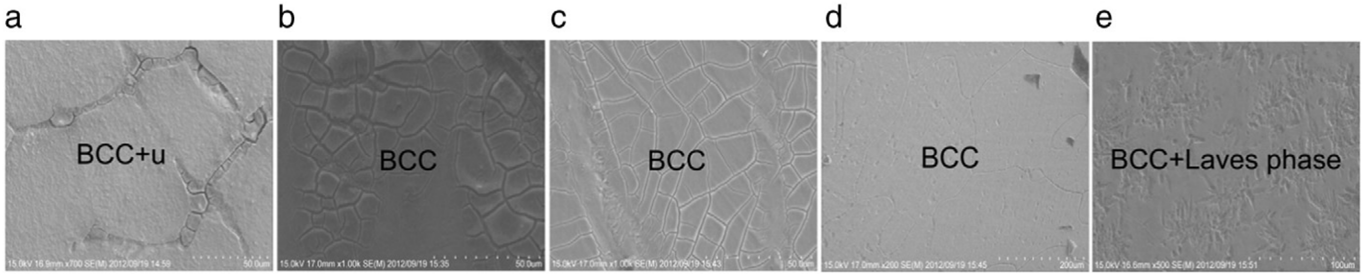

Fig. 2. SEM images of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$: (a) surface in the as-cast state, and (b) at 573 K, (c) at 773 K, (d) at 973 K, and (e) at 1173 K in as-annealed states.

preserves its original BCC phase while an unknown metastable phase formed upon casting dissolves in the BCC one and single-phase BCC structure is observed. At 1173 K a Laves V_2(Hf, Zr, Ti, Nb) solid solution phase precipitates from the BCC phase (Fig. 1(a)). The structure of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy in the as-solidified state has the BCC phase and two equilibrium Laves phases (Cr_2Nb and Cr_2Hf). This triple-phase structure is preserved and remained almost unchanged after annealing (Fig. 1(b)). The Laves phase formation is expected when the atomic size ratio of the largest and smallest elements in the composition exceeds 1.225 [11]. For the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy the R_{Hf}/R_V is 1.19 whereas the $R_{Hf}/R_{Cr} = 1.25$ for $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$, and it turns out that the formation of the Cr_2Hf phase is expected in these alloys.

Using the rule of mixtures (i.e. Vegard's law [28]) and the known concentrations of the alloying elements in the BCC phases, one can calculate the crystal lattice parameter a_{mix} of the respective disordered BCC solid solutions: $a_{mix} = \Sigma c_i a_i$, where c_i and a_i are the concentration and lattice parameter of the ith alloying element, respectively. Note that the lattice parameters of BCC Ti, Zr and Hf [12] are obtained via the thermal expansion. The lattice parameters of the BCC phase were determined to be $a_{mix} = 334.76$ pm for the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy and in the case of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy $a_{mix} = 332.04$ pm, whereas the lattice parameters determined by the XRD are 338.5 and 342.9 pm for the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloys, respectively. The results are listed in Table 1. The calculated value for the almost pure single-phase $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy is very close to the experimental result, whereas for the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy containing Laves phase the experimental value is seemingly larger than the calculated one.

Microstructure by SEM

Fig. 2 shows the SEM images of the surface of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ at as-cast state (a) and after annealing at 573 K (b), 773 K (c), 973 K (d) and 1173 K (e). The $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy preserves its original BCC phase while an unknown metastable phase formed upon casting dissolves in the BCC one and a single-phase BCC structure is observed. SEM images of the surface of the same alloy sample, after annealing at 1173 K, show typical cast dendrite and interdendrite structures (DR and ID). A Laves HfV_2 phase precipitates from the BCC phase. It can be deduced from the XRD results that the dendrite phase is the BCC solid solution structure while the interdendrite phase must be C15 HfV_2 intermetallic compound. It should be noted that the initial grain size of the BCC phase in as cast state (~13 μm) varies little by the annealing treatment at low temperatures (~13 μm at 573 K, ~12.5 μm at 773 K) but starts increasing at higher annealing temperatures (~69 μm at 973 and ~100 μm at 1173 K).

In the case of Cr containing alloy the Laves phase appeared in the as cast state already. The cast dendrite and interdendrite structures are shown through the chemical analysis results in Fig. 3. Note that the label Cr in Fig. 3(a) represents the Laves phase. The results of chemical analysis in as-cast and annealed alloys are given in Table 2. It is found in the case of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ BCC phase is enriched in Ti, Zr, Hf, and Nb and depleted in Cr. During solidification the Cr enriched in

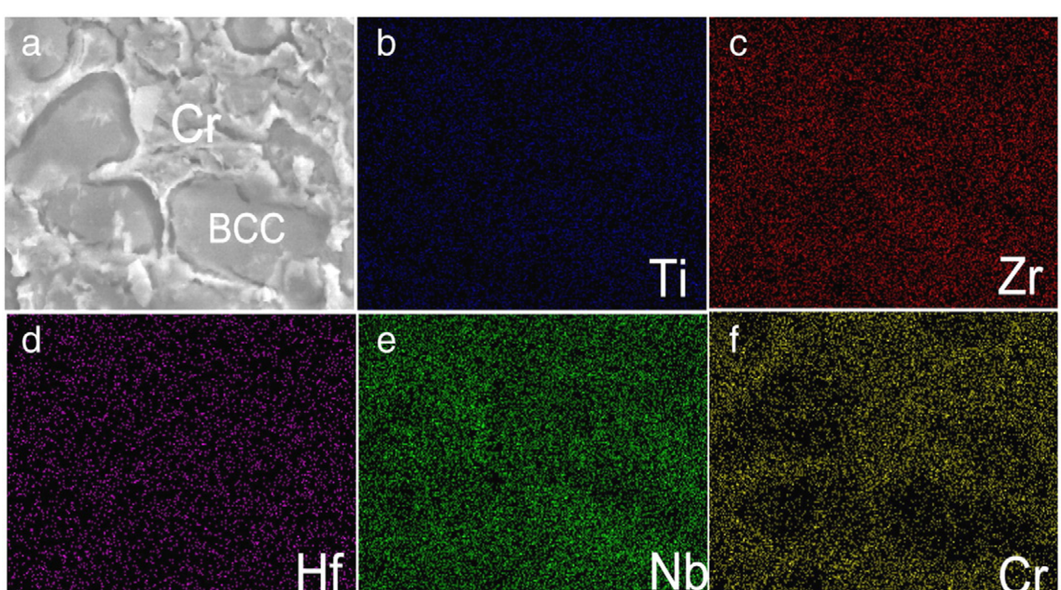

Fig. 3. EDS maps of alloying components in the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ HEA after annealing at 1173 K. In (a) the label Cr stands for the Laves phase.

Table 2
Quantitative chemical analysis of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr) samples in as cast (T = 298 K) and annealed (T = 1173 K) states.

HEA	T (K)	Phase	Composition	Ti	Zr	Hf	Nb	V	Cr
X = V	298	BCC phase	(at.%)	19.42	20.36	20.47	24.32	15.43	–
		Grain boundaries	(at.%)	20.28	21.12	18.5	22.54	17.56	–
	1173	BCC phase	(at.%)	19.17	19.33	22.58	22.13	16.8	–
		Grain boundaries	(at.%)	20.51	19.18	23.73	22	14.58	–
X = Cr	298	BCC phase	(at.%)	18.44	21.9	24.26	26.29	–	9.11
		Grain boundaries	(at.%)	18.83	19.92	22.49	15.74	–	23.01
	1173	BCC phase	(at.%)	18.14	19.96	23.03	20.76	–	18.11
		Grain boundaries	(at.%)	14.46	16.24	21.69	15.00	–	32.61

grain boundary regions helps in the formation of the Cr-rich FCC particles (Laves phases of C15 type: Cr_2Nb and Cr_2Hf) detected by XRD (see Fig. 1(b)).

Mechanical properties

The compression test result (Fig. 4) of the samples heat-treated for short time (10 min) at different possible working temperature peaks shows practically overlapping stress–strain curves, similar to the hardness result (Table 3). This result suggests that these relatively low temperatures and short time annealing produced no substantial phase precipitation in accordance with the XRD graphs (Fig. 1). Table 3 lists the Young's modulus *E*, yield stress σ_y, fracture strength σ_{max}, plastic strain ε_p and Vickers hardness $HV_{0.1}$ for the as-cast and heated $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr) alloys.

Fig. 4(a, c) shows the engineering stress (σ) as function of strain ε and Fig. 4(b, d) is the true stress versus curves of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloys obtained during compression testing for as-cast and heat treated alloys. The σ_y values given in Table 3 for the as-cast samples were 1170 and 1375 MPa, respectively. The ductility of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr) indicated the plastic strain listed in Table 3 and the stress versus strain plotted in Fig. 4. It should be noted that the large ε_p for the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy exceeds 30% during compression test (see Fig. 4(a) and Table 3). Some plastic deformability is observed in the case of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy as well (ε_p ~ 2–4%), although its yield strength is much higher than that of other alloys. The relative large volume fraction of Laves phase limits ductility of Cr-containing alloy. Under similar ductility, the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy exhibits higher strength than that (σ_y = 929 MPa) of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Ta_{20}$ [4]. For $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy, the σ_y shown in Table 3 is about 1322–1457 MPa at different temperatures. At room temperature, the σ_y of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ is relatively large, compared to the available values of refractory HEAs (1058 MPa for TaNbWMo and 1246 MPa for TaNbWMoV [3]).

Table 3 shows the Vickers hardness $HV_{0.1}$ of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr) alloys as a function of the annealing temperature. The

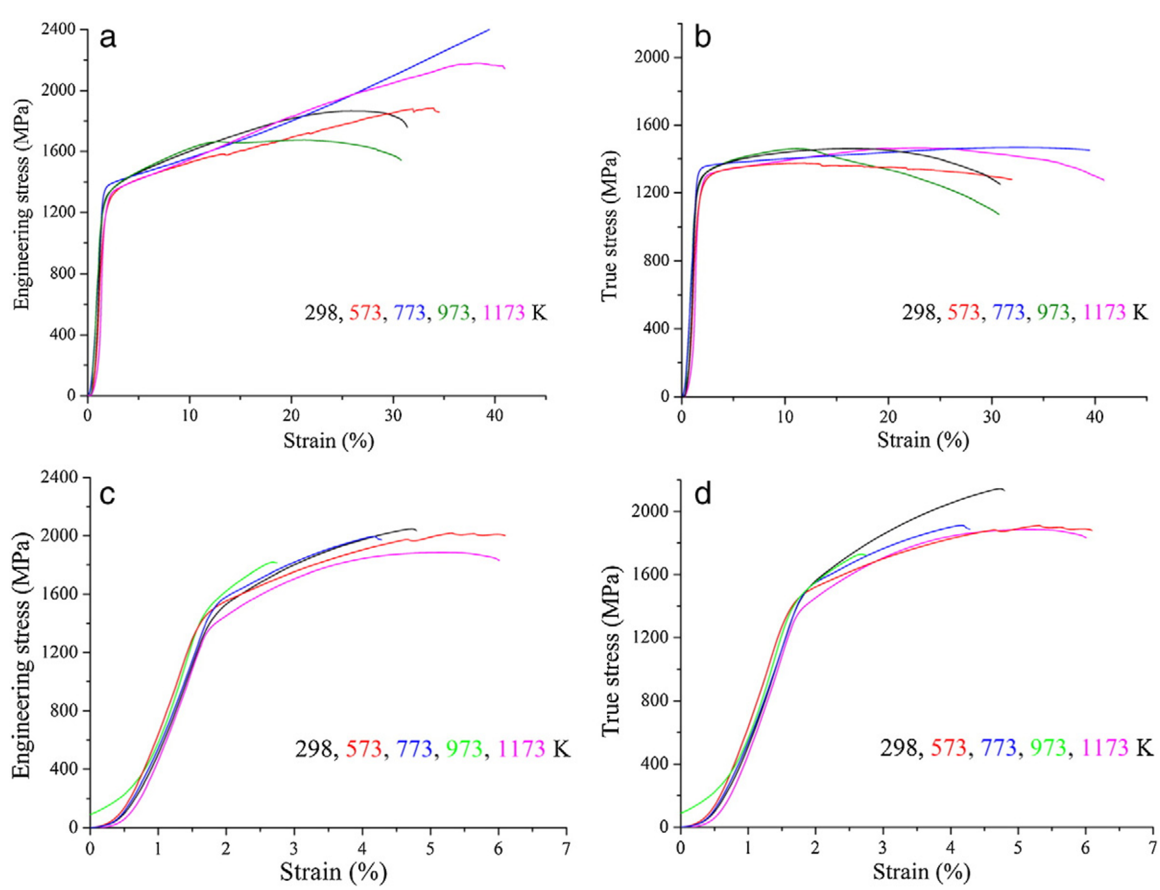

Fig. 4. Engineering stress and true stress vs. strain compression curves for the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ (a, b) and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ (c, d) alloys in the as-cast and annealed states.

Table 3
Room-temperature compressive mechanical properties for $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X$ (X = V, Cr) as-cast and heat-treated alloys, namely Young's modulus (E), yield stress (σ_y), fracture strength (σ_{max}), plastic strain (ε_p) and Vickers hardness at 1 N load ($HV_{0.1}$).

Alloy	Microstructure	$T_{annealing}$ (K)	E (GPa)	σ_y (MPa)	σ_{max} (MPa)	ε_p (%)	$HV_{0.1}$ (GPa)
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$	BCC + u	298	128	1170	1463	29.6	3.8
	BCC	573		1120	1376	30	3.85
	BCC	773		1253	1393	38	3.82
	BCC	973		1140	1463	29.7	3.86
	BCC + V_2M	1173		1157	1468	39.6	3.81
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$	BCC + L1 + L2	298	112	1375	2130	2.8	4.55
	BCC + L1 + L2	573		1420	1885	4.2	4.57
	BCC + L1 + L2	773		1457	1908	2.2	4.70
	BCC + L1 + L2	973		1322	1726	1.4	4.80
	BCC + L1 + L2	1173		1328	1884	4.5	4.50
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}Ta_{20}$ [3]	BCC	296	–	929	–	50	3.83
$W_{20}Nb_{20}Mo_{20}Ta_{20}$ [4]	BCC	296	–	1058	1211	1.5	4.46
$W_{20}Nb_{20}Mo_{20}Ta_{20}V_{20}$ [3]	BCC	296	–	1246	1270	1.7	5.25

Herein u is an unknown compound, L represents Laves phase.

statistical error is ±0.1 GPa. The $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy exhibits the higher Vickers hardness values than $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$.

The samples after compression in as-cast state and after annealing at 1173 K as well as their magnified fracture surfaces are shown in Figs. 5, 6, 7 and 8.

Figs. 5(a, b, c) and 6(a, b, c) are about the samples of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy before and after annealing at 1173 K. The alloy was deformed to 30% compression strain without fracture which can be seen in Fig. 5(b) while after annealing at 1173 K some cracks appear in Fig. 6(b). That is confirming good plasticity of this material in as cast state. Figs. 5(c) and 6(c) show the deformed grains of these samples that were elongated in the radial direction.

The fracture surface corresponds to ductile inter-crystalline fracture when propagation of the crack took place through the grain body.

The ductile fracture leads to the formation of dimples of different sizes and morphology, while the brittle fracture occurs by quasi-cleavage of the Laves phases. The brittle nature of the Laves phases leads to low ductility of the samples. A wide range of the mechanical properties is attainable through different combinations of these two phases: BCC phase as a relatively soft and ductile one while FCC phase (Laves phases) as hard but brittle phases.

In the case of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$, Figs. 7(a) and 8(a) show cast dendrite and interdendrite structures. It can be deduced from the XRD results that the dendrite phase must be the BCC solid solution structure while the interdendrite structures consist mostly of the two Laves phases.

For this alloy ductile (Fig. 7(c)) and brittle (Fig. 8(c)) regions are seen after the compression test on the fracture surfaces. Fig. 7(c) shows dimple fracture of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ as cast sample. The sizes of dimples are about 5–10 µm. The large depth of the cavities of the relief shows that the work of destruction of the samples, despite on the presence of a large proportion of the fragile intermetallic phase component, is significant. After annealing at 1173 K this alloy sample shows dimpled fracture. Fracture energy of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy samples was determined to be 10.4 kJ/cm^3 and 54.4 kJ/cm^3, respectively.

Theoretical calculations

Table 4 shows the lattice parameters, elastic constants, and polycrystalline elastic moduli as well as the valence electron concentration (VEC) of the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr, Ta), $Ta_{25}Nb_{25}W_{25}Mo_{25}$, $Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$ and $Zr_{33.3}Hf_{33.3}Nb_{33.3}$ refractory HEAs. Vegard's law discussed above is often used to estimate the lattice parameter a_{mix} phase of alloys. Our ab initio predications are slightly larger than a_{mix} for the three refractory HEAs. Hence all alloys show a small but systematic positive deviation relative to Vegard's rule. We note that for all HEAs considered here, the calculated lattice parameter a_t is slightly larger than the experiments a_e, except for $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$. It is very likely that this deviation is due to the employed frozen-core approximation, which is known to yield somewhat larger lattice parameters as compared to all-electron calculations.

All HEAs considered here are predicted to be mechanically stable according to the dynamical stability conditions $c_{44} > 0$, $c_{11} > |c_{12}|$ and $c_{11} + 2c_{12} > 0$. For the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr, Ta) refractory HEAs, the polycrystalline elastic moduli are very similar. It is found that V has a small effect on B, G and E of $Ta_{25}Nb_{25}W_{25}Mo_{25}$ alloys.

According to Tables 3 and 4, our measured Young's modulus is 128 (112) MPa for $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ ($Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$) whereas the ab initio E is 95 (104.4) MPa. It should note that $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ is almost a single BCC phase whereas $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ consists of a mixture of BCC and Laves phases. The theoretical results are expected to be valid for the completely random and homogeneous BCC solid

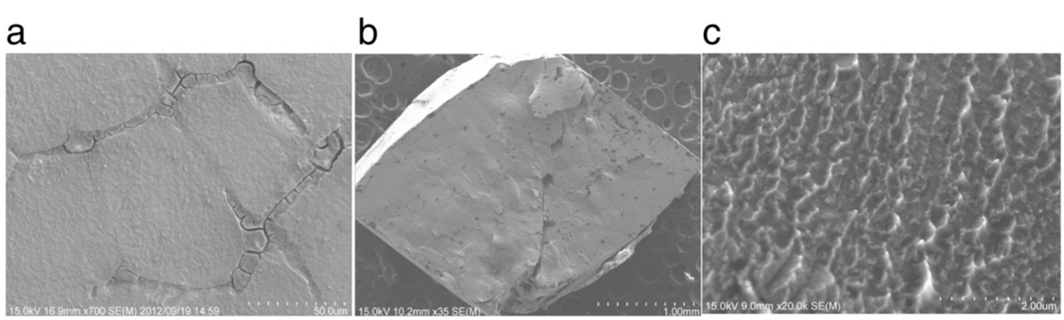

Fig. 5. SEM images of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ in the as-cast state: (a) surface, (b) fracture macroscopic, and (c) fractured microscopic.

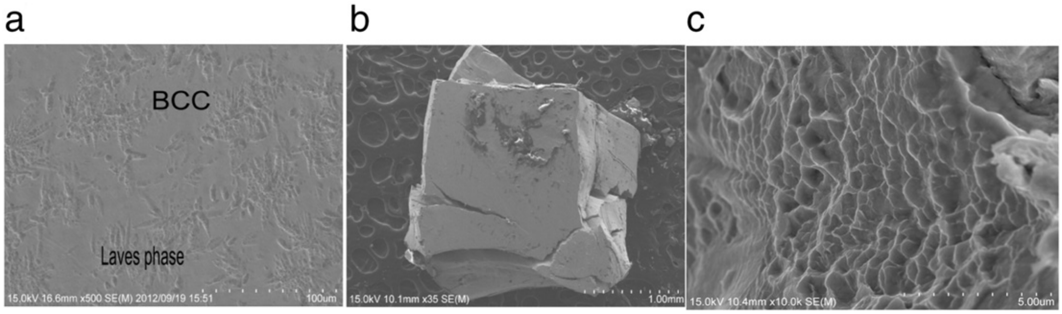

Fig. 6. SEM images of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ after annealing at 1173 K: (a) surface, (b) fractured macroscopic, and (c) fractured microscopic.

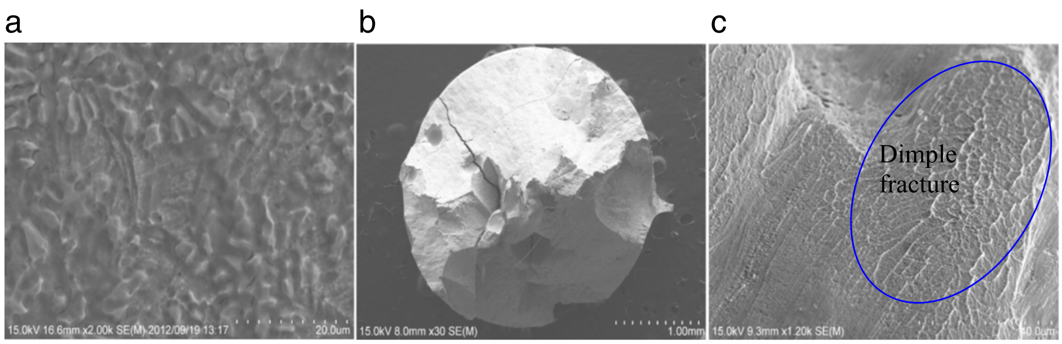

Fig. 7. SEM images of the surface (a) and fractured: macroscopic (b) and microscopic (c) of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ in as-cast state.

solutions. The elastic anisotropy of cubic lattices is often expressed in terms of Zener ratio $A_Z = c_{44}/c'$. The ratio A_{VR} can be calculated according to the Voigt and Reuss bounds [27], which is an alternative measure of the elastic anisotropy. In order to illustrate the elastic anisotropy of refractory HEAs, in Fig. 9 we plot the three dimensional E for $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr, Ta), $Zr_{33.3}Hf_{33.3}Nb_{33.3}$, $Ta_{25}Nb_{25}W_{25}Mo_{25}$, and $Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$. For a fully isotropic material, the tetragonal shear modulus $c' = (c_{11} - c_{12})\,/\,2$ equals the cubic shear modulus c_{44}, so we have $A_Z = 1$ and $A_{VR} = 0$. Recently the valence electron concentration (VEC) was used to study the isotropy of the refractory alloys [25, 29]. For $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr, Ta) and $Zr_{33.3}Hf_{33.3}Nb_{33.3}$, $A_Z > 1$ and VEC < 4.7, the E along [111] direction is larger than that along two other common directions [001] and [110], whereas the opposite is for $Ta_{25}Nb_{25}W_{25}Mo_{25}$ and $Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$ alloys with the $A_Z < 1$ and VEC > 4.7.

The calculated large positive Cauchy pressure (c_{12}–c_{14}) suggests that these refractory HEAs have strong metallic character and enhanced ductility. According to Pugh [30], materials with B/G ratio above 1.75 are ductile. For isotropic materials, the Pugh criteria for ductility imply $v > 0.26$, which has been confirmed for bulk metallic glasses [31]. Note that the criterions are derived from the single-phase solid solutions. All values in Table 4 indicate enhanced ductility for these refractory alloys. In fact the high ductility has been reported for single-phase $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Ta_{20}$ while the limited ductility at room temperature and extensive compressive plasticity above 873 K show for the single-phase $Ta_{25}Nb_{25}W_{25}Mo_{25}$ and $Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$ alloys [3,4]. Whereas the limited ductility $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy may result from the large volume fraction of Laves phase.

Conclusions

Two new refractory high-entropy alloys, $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$, were produced by induction melting and casting. The $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ has a BCC + unknown intermetallic phase structure while $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ alloy forms the Cr_2Nb and Cr_2Hf intermetallic compounds between the BCC phase particles. A single

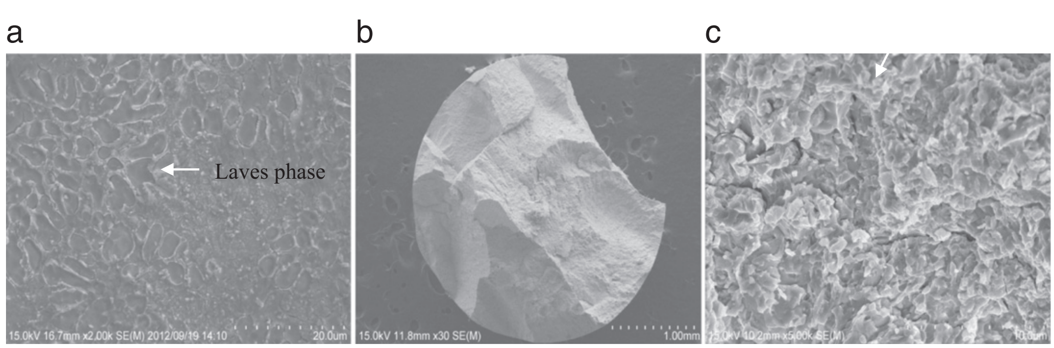

Fig. 8. SEM images of $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ after annealing at 1173 K: (a) surface, (b) fractured macroscopic, and (c) fractured microscopic.

Table 4
Theoretical lattice parameters and elastic properties of refractory HEAs.

HEAs	a_t	a_{mix}	a_e	B	c_{11}	c_{12}	c_{44}	c'
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$	340.1	341.3	338.5	126.6	149.5	115.1	55.0	17.2
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$	337.1	337.9	342.9	117.2	153.1	99.3	49.2	26.9
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}Ta_{20}$	345.7	348.2	340.4	136.3	160.2	124.4	62.4	17.9
$Ta_{25}Nb_{25}W_{25}Mo_{25}$	326.9	330.1	321.3	261.6	413.5	185.6	69.0	114
$Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$	323.2	324.7	318.3	245.1	380.8	177.3	61.2	102
$Zr_{33.3}Hf_{33.3}Nb_{33.3}$	356.3	358.6	348.7	113.2	127.5	106.1	57.1	10.7

	c_{44}/c'	$c_{12}-c_{44}$	G	E	ν	B/G	A_{VR}	VEC
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$	3.200	60.1	34.6	95.0	0.375	3.663	0.154	4.4
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$	1.827	50.1	38.6	104.4	0.352	3.034	0.043	4.6
$Ti_{20}Zr_{20}Hf_{20}Nb_{20}Ta_{20}$	3.491	62.0	37.9	104.1	0.373	3.596	0.176	4.4
$Ta_{25}Nb_{25}W_{25}Mo_{25}$	0.605	116.6	84.4	228.7	0.354	3.097	0.030	5.5
$Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$	0.602	116.1	75.1	204.5	0.361	3.263	0.031	5.4
$Zr_{33.3}Hf_{33.3}Nb_{33.3}$	5.351	49.01	29.7	81.89	0.379	3.815	0.298	4.3

a represents the lattice parameter of BCC structure (in units of pm), c_{11}, c_{12}, and c_{44} are the three independent elastic constants, $c_{12}-c_{44}$ is the Cauchy pressure, *B*, *G* and *E* are the bulk, shear and Young's moduli (in units of GPa), ν is the Poisson ratio and *B*/*G* is the Pugh ratio. a_t, a_{mix}, and a_e stand for the theoretical, estimated (using Vegard's rule) and experimental lattice parameters, respectively.

BCC phase was formed in the $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy as a result of annealing at 573, 773, and 973 K, whereas a V_2Hf phase precipitated from the BCC phase at 1173 K.

For both alloys almost no changes were observed in the Vickers microhardness before and after annealing. The $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ and $Ti_{20}Zr_{20}Hf_{20}Nb_{20}Cr_{20}$ as-cast alloys have the high compression yield strength σ_y values of 1170 and 1375 MPa, respectively, at room temperature. The $Ti_{20}Zr_{20}Hf_{20}Nb_{20}V_{20}$ alloy shows considerable strengthening and homogeneous deformation. Replacement of V with Cr produced strengthening of the alloy due to Laves phase precipitation in the BCC matrix. Our experiments and ab initio calculations indicate that these alloys are relatively ductile.

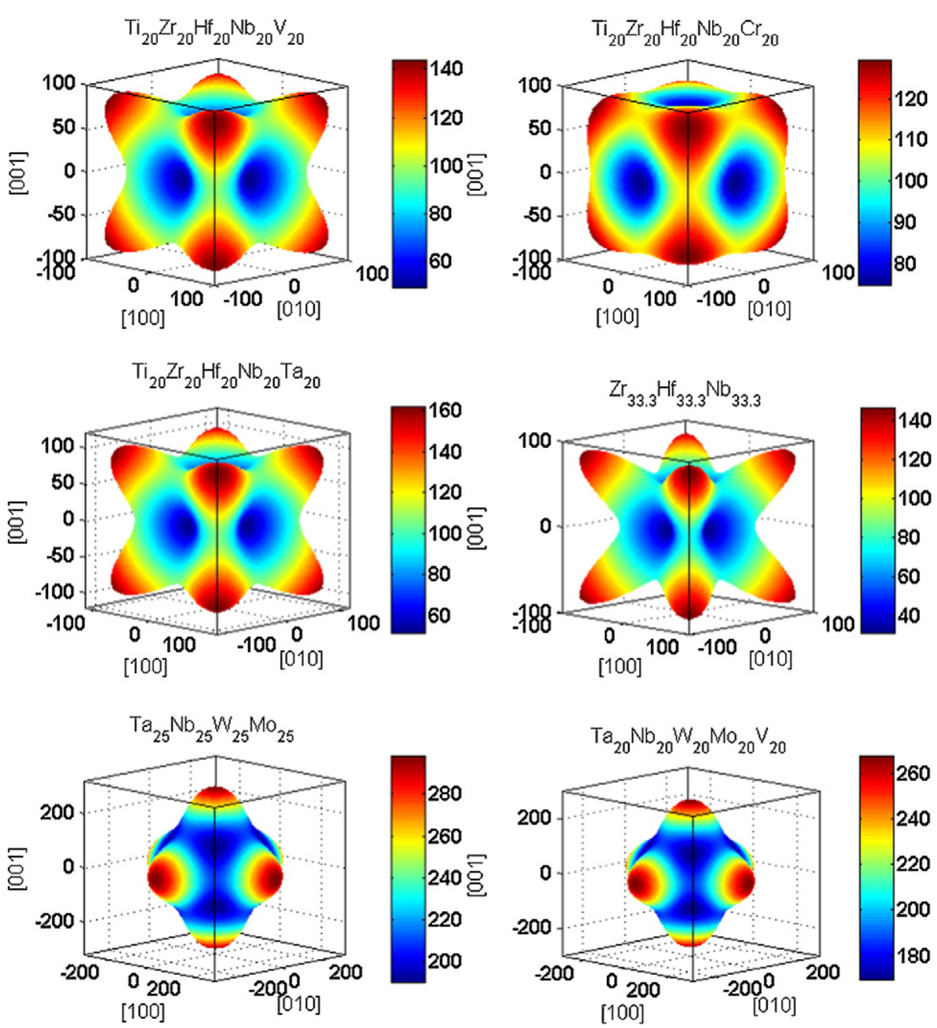

Fig. 9. Characteristic surfaces of the Young's modulus for $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V, Cr Ta), $Zr_{33.3}Hf_{33.3}Nb_{33.3}$, $Ta_{25}Nb_{25}W_{25}Mo_{25}$, and $Ta_{20}Nb_{20}W_{20}Mo_{20}V_{20}$. The values on the color scale and on the axes are in GPa.

The $Ti_{20}Zr_{20}Hf_{20}Nb_{20}X_{20}$ (X = V or Cr) alloys show a large elastic and plastic strain limits while the BCC phase is stable against heating for 600 s up to 1173 K.

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